12TH BIOLOGY · CHAPTER 13 · CHAPTER TEST

Thermoregulation and Osmoregulation

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Biology	**Total Marks:** 26
Chapter: 13	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. Positive feedback loops are less common because they:

(a) Reverse the stimulus (b) Work without receptors
(c) Use no effectors (d) Enhance the original stimulus

2. Which animal is an osmoconformer?

(a) Kangaroo rat (b) Hagfish
(c) Freshwater fish (d) Sparrow

3. Chloride cells of marine fishes are present in the:

(a) Gills (b) Rectum
(c) Kidneys (d) Skin

4. Which group of animals is uricotelic?

(a) Sponges (b) Mammals
(c) Birds (d) Protozoans

5. Humans are:

(a) Ectothermic poikilotherms (b) Endothermic homeotherms
(c) Ectothermic homeotherms (d) Endothermic poikilotherms

6. Which response occurs when body temperature falls below 37 °C?

(a) Vasodilation (b) Sweating
(c) Vasoconstriction (d) Decrease in thyroxine

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. How do control centres help to maintain homeostasis?
ii. Why do freshwater fishes rarely drink water but produce large volumes of dilute urine?
iii. How do marine fishes remove the excess salts from their bodies?
iv. Analyze why humans excrete uric acid even though they are not uricotelic.
v. Differentiate between ectotherms and endotherms.
vi. Why is thermoregulation necessary for the working of enzymes and cell membranes?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Explain the negative and positive feedback mechanisms of homeostasis with one example of each.
(b) Describe the regulatory strategies in man for thermoregulation.

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